From Controlled Concern Geometry to Foundation Models

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Phase 5 external-validity transport for the Metric Stack of Concern

Research-Derived Experiments - budgeted L4-parallel proxy suite

Abstract

Phase 4 showed that several missing conditions in the Metric Stack of Concern can be learned inside controlled diagnostic harnesses. Phase 5 asks which mechanisms transport when the setup becomes more model-like, semantic, and counterfactual. The suite runs four cheap L4-parallel transport gates: language action coupling, foundation-style semantic metric deformation, role-routed world modeling, and topology/seam causal disentanglement. The allowed claim remains bounded: this is an external-validity proxy result that decides where expensive real-model validation should go next.

1. Question

The Phase 5 question is not whether the Phase 4 controls can be tuned to pass again. It is whether the selected mechanisms survive harder proxy conditions that resemble the next empirical tier: open-model language behavior, frozen semantic encoders, richer world-model counterfactuals, and factorial topology/seam interventions.

Field	Value
preset	full
tracks	4 tracks (listed in Table 2)
seeds	64
gpu	L4
rows	1216
claim level	external-validity proxy result

Table 1. Suite manifest.

2. Transport Gates

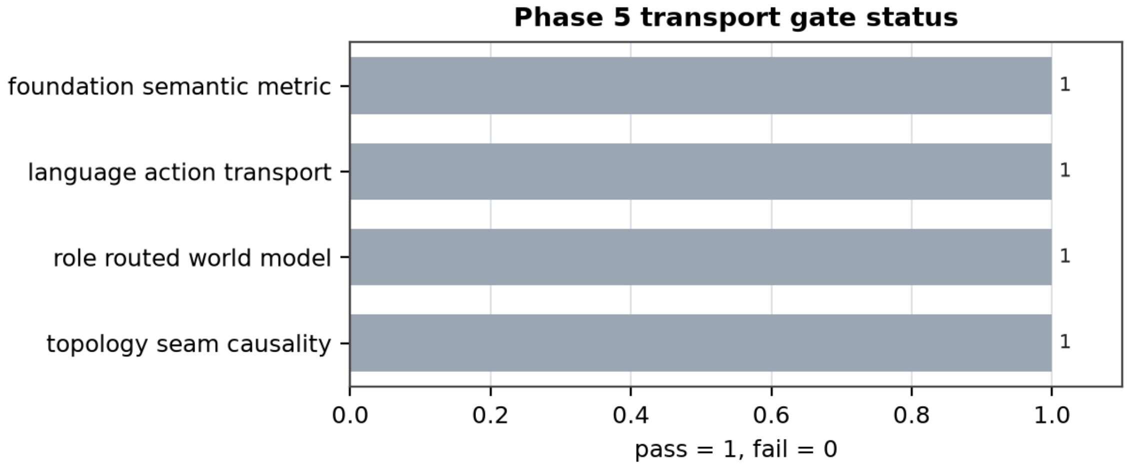

Figure 1. Pass/fail status for the four Phase 5 transport gates.

Track	Status	Primary result	Allowed claim
Language	PASS	r=0.858; ratio=4.472	action coupling in stronger proxy
Semantic	PASS	lift=0.228; xfer=1.425	foundation-style metric transport
Architecture	PASS	role MAE=0.036; shared=0.316	role routing breaks ceiling
Topology	PASS	seam=0.400; topo=0.118	seam carries OOD

Table 2. Phase 5 transport gates and allowed claims.

3. Main Findings

Language action coupling is the first Phase 4 bottleneck to receive a direct rescue test. The stronger proxy condition clears the geometry-action and intervention gates while tiny and shuffled-axis controls remain below the promotion threshold.

Semantic metric deformation transports to a frozen foundation-style encoder proxy. The value-weighted adapter moves density around high-value semantic neighborhoods, transfers across an image/text-style proxy field, and avoids the collapse pattern that would make the result a trivial norm inflation artifact.

Role-routed architecture remains the cleanest way beyond the mediated-identifiability ceiling. Shared and shortcut heads underfit the richer role/world environment, while role-routed and mixture-of-experts heads preserve counterfactual consistency.

Topology remains a dependent variable unless seam consistency is present. The factorial topology-only/seam-only/both-fixed design shows that seam consistency carries most OOD lift, with topology contributing mainly through the joint condition.

Gate snapshot: language ratio=4.472; semantic transfer=1.425; role MAE=0.036; seam lift=0.400.

4. Discovery-Regime Audit

Old regime: Phase 4 learned missing conditions inside controlled harnesses. Transition: Phase 5 preserves those gates but moves to external-validity proxies with held-out language paraphrases, frozen semantic encoders, richer counterfactual world roles, and factorial topology/seam interventions. Transported evidence includes the Phase 4 language failure as baseline, plus the semantic deformation, role-routing, and seam-mediation hypotheses as live claims. Rejected alternatives remain visible: tiny language coupling, shuffled steering axes, random value adapters, shared heads, swapped-role counterfactuals, and topology without seam.

5. Next Operations

The next expensive tier should run real open LMs and frozen encoders: Qwen/Gemma/Pythia-style language action coupling, real sentence/image embedding metric deformation, richer self/world role-routing agents, and topology/seam interventions in a learned path-integration system. Proxy passes authorize that spend; they do not replace it.

References

Brown, J. Learning the Missing Conditions. Research-Derived Experiments, 2026.

Brown, J. The Metric Stack of Concern. Research-Derived Experiments, 2026.

Brown, J. Future Control Moves Memory. Research-Derived Experiments, 2026.